$\mathbf{EFX_0}$ Allocations with at Most Three Relevant Goods

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Abstract. An allocation of indivisible goods is EFX_0 if no agent envies another agent's bundle after removing any single good from it, even a good that the envious agent values at zero. We prove that every instance with nonnegative additive valuations in which each agent positively values at most three goods has a complete EFX_0 allocation in which all bundles but at most one have at most two goods. An algorithm we call Draft and Exchange (DE) computes one in time $O(n(n+m))$. Agents that can leave with at most one good do so; the others draft at most one good each; then, until a free agent can take all the leftover goods (or every agent holds its pair), goods are traded along a chain or a ring of agents, each of which gets strictly better off. There are at most $4n$ trades. The key fact is that when no free agent can take the leftovers, the free agents have enough distinct obstacles to point at, and following these pointers closes a ring. The proof is machine-checked in Lean, together with the algorithm, its bound of $4n$ trades and its running time.

Keywords: Fair division · Indivisible goods · EFX · Serial dictatorship · Trading cycles

1 Introduction

We allocate a set M of m indivisible goods to a set N of $n \geq 1$ agents. Agent i has an *additive valuation*: a value $v_i(g) \geq 0$ for each good g, and $v_i(S) = \sum_{g \in S} v_i(g)$ for every set S of goods. An *allocation* X gives every good to exactly one agent; X_i is agent i's *bundle*. Every allocation in this paper is *complete*: no good may be thrown away.

Envy-freeness can be impossible with indivisible goods (one good, two agents who value it). *Envy-freeness up to any good* weakens it [4]: the envy of i for X_j must disappear as soon as any single good is removed from X_j. Two versions differ in which goods may be removed:

$$\begin{aligned} &\text{EFX:} && v_i(X_i) \geq v_i(X_j \setminus \{h\}) && \text{for all } i \neq j \text{ and } h \in X_j \text{ with } v_i(h) > 0; \\ &\mathrm{EFX}_0\text{:} && v_i(X_i) \geq v_i(X_j \setminus \{h\}) && \text{for all } i \neq j \text{ and } h \in X_j. \end{aligned}$$

EFX_0 implies EFX, and the two coincide when all values are positive. Whether every instance with additive valuations has a complete EFX

allocation is a central open question of fair division [4]. Under EFX_0, if another agent's bundle contains a good that agent i values at 0, then i must not envy that bundle at all, since removing that good does not lower its value to i.

Example 1 (EFX, but not EFX_0). Two agents share goods a, b, c. Agent 1 values a, b, c at 3, 2, 0; agent 2 values only c, at 1. The allocation $X_1 = \{b\}$, $X_2 = \{a, c\}$ is EFX: agent 1 may remove only a from X_2, and $\{c\}$ is worth $0 \leq 2$; agent 2 values $\{b\}$ at 0. It is not EFX_0: removing c from X_2 leaves $\{a\}$, worth $3 > 2$ to agent 1. The allocation $X_1 = \{a\}$, $X_2 = \{b, c\}$ is EFX_0.

1.1 Results

Write $R_i = \{g \in M : v_i(g) > 0\}$ for the goods agent i values, its *relevant* goods. If every agent values at most two goods, serial dictatorship works: the agents, in any order, each take a favourite remaining good, and the last agent takes everything left ([12]; Corollary 1 below). At three goods this breaks. An agent that values a, b, c at 4, 3, 2 may take a and then see b and c end up in the last agent's bundle together with a third good; removing that third good leaves b and c, worth $5 > 4$.

Theorem 1 (existence). *Every instance with $n \geq 1$ agents and nonnegative additive valuations in which every agent positively values at most three goods has a complete EFX_0 allocation.*

Theorem 2 (one large bundle). *Every such instance has a complete EFX_0 allocation in which all bundles but at most one have at most two goods.*

Theorem 3 (algorithm). *Algorithm Draft and Exchange (*DE*, Section 6) computes such an allocation for every instance in which every agent positively values at most three goods, with at most n peeling steps and at most $4n$ trades, in time $O(n(n+m))$. On real values, every decision of* DE *compares two sums of at most two values of one agent.*

1.2 What is new

We ask how many goods each agent may value before complete EFX_0 allocations become hard to find, with nothing else restricted: any number of agents and goods, any number of agents valuing the same good, and any nonnegative additive values. If every agent values at most two goods,

serial dictatorship works [12]. For three goods, no earlier result settles the question, as far as we know.[1] Our contributions are:

- *Existence* (Theorem 1): the first proof that complete EFX_0 allocations exist when every agent values at most three goods.
- *Shape* (Theorem 2): all bundles but at most one have at most two goods. This cannot be improved in the obvious ways: some instances need a bundle of three goods, and some need four (Appendix B). We know of no shape result of this kind for EFX or EFX_0.
- *Algorithm* (Theorem 3): DE computes such an allocation in time $O(n(n+m))$, with at most $4n$ trades.
- *A complete, machine-checked proof*: every result of this paper is checked in Lean (Section 8).

Why known results do not settle it. Viswanathan and Mehta [16] study the same restriction, with up to four goods per agent, but for ordinary EFX, and give only a proof sketch in a three-page extended abstract. EFX does not give EFX_0, and the difference matters exactly here: when agents value few goods, most goods are worth 0 to most agents, and under EFX_0 such goods still count (Example 1); their algorithm gives the goods nobody values to an arbitrary agent, which can break EFX_0. Conversely, since EFX_0 implies EFX, Theorem 1 is a complete proof of the three-goods case of their result. Every other existence result for complete EFX_0 allocations that we know of restricts something our model leaves free: the number of agents [6,3], the number of goods compared with the number of agents [14,5], the number of agents that value a good [1,7,15,9], the girth of the hypergraph whose edges are the goods [10], or the number of distinct valuations [8]. For example, six agents valuing $\{g_0, g_1, g_6\}$, $\{g_0, g_1, g_7\}$, $\{g_0, g_2, g_8\}$, $\{g_2, g_3, g_4\}$, $\{g_3, g_5, g_9\}$ and $\{g_4, g_5, g_{10}\}$, with pairwise distinct values between 1 and 2, form an instance of our model that none of these results covers: $m = 11 = n + 5$; three agents value g_0; the first two agents share two goods, so the hypergraph has girth 2; and the six valuations differ. Finally, Theorem 1 would follow from the conjecture that complete EFX allocations always exist for additive valuations, by a perturbation argument [6], but that conjecture is open for four or more agents [3].

[1] Literature search of 25 September 2026: arXiv (92 queries; the full texts of the 165 papers from April 2024 on that they returned), the web, Zenodo, Google Scholar, Crossref and the EC 2026 programme, recorded in `proofs/novelty.md` of [11].

1.3 How the proof works

Peeling. If an agent values its favourite remaining good at least as much as all its other remaining goods together, it takes that good and leaves; nobody envies a bundle of one good after removing it. When nobody can leave, every remaining agent values exactly three remaining goods $a \succ b \succ c$, with $v(a) < v(b) + v(c)$.

The draft. The remaining agents, in turn, each take the best of their three goods still available, or nothing. Afterwards every good that an agent ranks above what it holds, a good it *wants*, is held alone by another agent. We keep this property, *validity*, throughout. The goods nobody took are *leftover*.

Finishing. To finish, one agent takes all the leftover goods. It must be *free*: nobody may want its good. The only agents that could still envy its big bundle are its *blockers*: agents holding their top good a whose b and c would both be in it. Each blocker is fixed by moving one of these two goods, which is leftover, to another free agent, one good each. So a free agent can finish when its blockers' leftover goods are at most as many as the other free agents (Lemma 7 and Theorem 4).

Improving. If no free agent can finish, some agents can be made better off. If an agent holding its top finds its b and c both leftover, it takes them, and its top goes to an agent that wants it, and so on: a *chain*. Otherwise every free agent failed to finish because it has at least as many blockers with distinct leftover goods as there are free agents, so each free agent can pick a blocker whose leftover good no other free agent picked. Now every agent points at someone who would gladly take its good: a free agent at its blocker, which completes its pair with that good and its leftover good, and any other agent at an agent that wants its good. Following the pointers closes a *ring*, and passing the goods one step along it makes everyone on it better off (Theorem 5). Scores only grow, so this ends, with a free agent that can finish or with every agent holding its pair. Section 6 ends with a small example, and Section 7 has a larger one.

Status. The proofs below are written proofs, and they are machine-checked in Lean 4, together with the algorithm, its bound of $4n$ trades, its running time, and, through natural-number values that answer the same comparisons, its use on real values (Section 8). They were also checked by computer on millions of states. The reviews so far were by AI agents (Section 8); there has been no human peer review.

Related work. For additive valuations, complete EFX allocations are known to exist only in special cases, listed above and compared in Table 1

(Appendix C); for monotone valuations that are not additive, they need not exist [2,13]. All of these results, ours included, use the all-goods form EFX$_0$, except Viswanathan and Mehta's [16]. For at most two goods per agent, serial dictatorship has been formalized in Lean [12].

2 Peeling

Agents and goods are numbered, and "index order" and "the first" refer to these numbers. A good is *alone* in an allocation if it is the only good of its bundle.

Lemma 1 (peeling). *Let $i \in N$ and $P \subseteq M$ with (i) $v_i(P) \geq v_i(M \setminus P)$ and (ii) $|P| \leq 1$. If the instance with agents $N \setminus \{i\}$ and goods $M \setminus P$ has an EFX$_0$ allocation Z, then Z together with $X_i = P$ is an EFX$_0$ allocation of the whole instance.*

Proof. Agent i and a bundle Z_j: $v_i(Z_j \setminus \{h\}) \leq v_i(Z_j) \leq v_i(M \setminus P) \leq v_i(P)$. Agent $j \neq i$ and the bundle P: removing its only good leaves nothing. Agent $j \neq i$ and a bundle Z_k: this is a condition of Z. □

Rule R1. Let G be the set of remaining goods, and p agent i's favourite good of G (largest value; between equal values, the lower index). If $G = \emptyset$ or $v_i(p) = 0$, so that i values no remaining good, then i may leave with $P = \emptyset$. If $v_i(p) > 0$ and $v_i(G \setminus \{p\}) \leq v_i(p)$, then i may leave with $P = \{p\}$. Rule R1 applies, for instance, to every agent that values at most two remaining goods.

Corollary 1 (two relevant goods). *If $|R_i| \leq 2$ for every agent, then serial dictatorship, in any order of the agents, each taking a favourite remaining good (or nothing once no good is left), with the last agent taking all remaining goods, returns an EFX$_0$ allocation.*

Proof. By induction on the number of agents. One agent faces no condition. Otherwise the first agent takes a favourite remaining good p, or nothing, and values at most one other remaining good, worth at most $v_i(p)$; so its bundle satisfies conditions (i) and (ii) of Lemma 1, which reduces to the smaller instance. □

When $|R_i| = 3$ we name agent i's relevant goods a_i, b_i, c_i so that $v_i(a_i) \geq v_i(b_i) \geq v_i(c_i) > 0$, breaking ties in favour of the lower index. This gives a strict *ranking* $a_i \succ_i b_i \succ_i c_i$, and $g \succ_i h$ implies $v_i(g) \geq v_i(h)$. Agent i is *strictly balanced* if $v_i(a_i) < v_i(b_i) + v_i(c_i)$.

Lemma 2 (when nobody can leave). *Let agent i value at most three goods of the set G of remaining goods, and suppose that rule R1 does not apply to i. Then i values exactly three goods of G and is strictly balanced with respect to them.*

Proof. Rule R1 fails, so $v_i(p) > 0$ and $v_i(G \setminus \{p\}) > v_i(p)$. If i valued at most two goods of G, then $v_i(G \setminus \{p\})$ would be the value of at most one good, at most $v_i(p)$. So i values exactly three goods of G, p is its top, and $v_i(b_i) + v_i(c_i) > v_i(a_i)$. □

In Sections 3–5 we are in the *core case*: every agent values exactly three goods and is strictly balanced. Goods that no agent values may exist.

3 States

Definition 1 (states and scores). *A* state *Y gives each agent i a* holding *Y_i: nothing, $\{a_i\}$, $\{b_i\}$, $\{c_i\}$, or its* pair *$\{b_i, c_i\}$, with the holdings pairwise disjoint. U is the set of agents holding their pair. The goods held by nobody are* leftover*; J is the set of leftover goods. Agent i's* score *is 4, 3, 2, 1 or 0 if it holds its pair, a_i, b_i, c_i or nothing. Since $v_i(a_i) < v_i(b_i)+v_i(c_i)$, a higher score never means a lower value.*

Definition 2 (wants, validity, free agents). *Let Y be a state. An agent $i \notin U$* wants *a good $g \in R_i$ if it holds nothing, or holds a good y with $g \succ_i y$. Agents in U want nothing. Y is* valid *if every good that some agent wants is held alone: it is the whole holding of some agent. An agent $i \notin U$ is* free *if it holds nothing or holds a good that nobody wants. F is the set of free agents.*

No agent wants its own good, so in a valid state the holder of a wanted good is another agent, outside U and not free. Leftover goods and the goods of pair holders are not wanted in a valid state. An agent outside $U \cup F$ holds a single good that some agent wants.

Lemma 3 (the draft). *Let the agents, in any order, each take the first of a_i, b_i, c_i not yet taken, or nothing if all three are taken. The result is a valid state with $U = \emptyset$.*

Proof. Each agent holds one of its goods or nothing, and no good is taken twice. If i wants g, then g was taken before i's turn, by an earlier agent as its only good, and holdings never change during the draft. □

Lemma 4 (staying valid). *Let Y be valid, and Y' a state in which every agent's score is at least its score in Y and every good wanted in Y is held alone. Then Y' is valid.*

Proof. Let i want g in Y'. If i holds nothing in Y', its score in Y was 0, so it held nothing in Y and wanted g there too. If i holds a single good y' in Y', its score in Y was below 4, so in Y it held nothing, or a good y with $y' = y$ or $y' \succ_i y$; since $g \succ_i y'$, again i wanted g in Y. So g is held alone in Y'. □

4 Finishing

Definition 3 (blockers, finishing). *Let Y be a state and o a free agent. A* blocker *of o is an agent $x \neq o$ that holds exactly its top, $Y_x = \{a_x\}$, and has $b_x, c_x \in J \cup Y_o$. Agent o* can finish *with a set $H \subseteq J$ if $|H| \leq |F| - 1$ and H contains b_x or c_x for every blocker x of o. Its* completion *X gives each good of H to a different free agent other than o, gives o the bundle $X_o = Y_o \cup (J \setminus H)$, and lets every other agent keep its holding (plus its good of H, if any).*

So a blocker of o is an agent that would find both b_x and c_x in o's bundle if o took all the leftover goods; H moves one of the two out of it. When every agent holds its pair, the completion with any agent o gives o the bundle $Y_o \cup J$ and lets the others keep their pairs.

Theorem 4 (soundness). *Let Y be a valid state, and X the completion with a free agent o that can finish with a set H, or, if every agent holds its pair, with any agent o. Then X is an EFX$_0$ allocation, and every bundle other than X_o has at most two goods.*

Proof. Sizes. A free agent other than o holds at most one good and receives at most one good of H; a pair holder holds two goods; every other agent keeps a single good.

Wanted goods stay alone. Let g be wanted, held alone by j. Then j is not free, so $j \neq o$ and j receives no good of H: $X_j = \{g\}$. Hence every good in a bundle of two or more goods is unwanted.

No envy. Let $i \neq j$ and $h \in X_j$. We show $v_i(X_j \setminus \{h\}) \leq v_i(X_i)$, where $v_i(X_i) \geq v_i(Y_i)$. If $|X_j| \leq 1$, then $X_j \setminus \{h\} = \emptyset$. Let $|X_j| \geq 2$, and $T = X_j \cap R_i$; it contains no good that i wants and none of i's holding.

- If i holds its pair, then $T \subseteq \{a_i\}$, and $v_i(T) \leq v_i(a_i) < v_i(b_i) + v_i(c_i)$.
- If i holds nothing, it wants all of R_i, so $T = \emptyset$.

– If $Y_i = \{y\}$, every good of T is ranked below y and worth at most $v_i(y)$. If $|T| \le 1$, then $v_i(X_j \setminus \{h\}) \le v_i(T) \le v_i(y)$. If $|T| = 2$, then $y = a_i$ and $T = \{b_i, c_i\}$. If $X_j = \{b_i, c_i\}$, removing h leaves one good, worth at most $v_i(b_i) \le v_i(a_i)$. Otherwise $|X_j| \ge 3$, so $j = o$ by the sizes, and $b_i, c_i \in X_o \subseteq Y_o \cup J$: i is a blocker of o. Then H contains b_i or c_i, which went to an agent other than o, a contradiction. (When every agent holds its pair, only the first case occurs.) □

5 Improving

Definition 4 (rings). *Let Y be a state. A* ring *is a cycle $w_0 \to w_1 \to \cdots \to w_{k-1} \to w_0$ of $k \ge 2$ distinct agents outside U, each holding a single good $y_t = y_{w_t}$, such that each arrow $w_t \to w_{t+1}$ (indices modulo k) is*

– *a* want arrow*: w_{t+1} wants y_t; or*
– *a* pair arrow*: w_t is free, w_{t+1} is a blocker of w_t, and the pair of w_{t+1} is $\{y_t, h_{t+1}\}$ with $h_{t+1} \in J$;*

and the leftover goods h_{t+1} of the pair arrows are pairwise distinct. Trading along the ring *gives each w_{t+1} the good y_t, together with h_{t+1} for a pair arrow, in place of its own good; agents off the ring keep their holdings.*

An arrow from a free agent is a pair arrow (nobody wants its good), and an arrow from any other agent is a want arrow.

Lemma 5 (ring). *Trading along a ring of a valid state gives a valid state in which every agent of the ring has a higher score and every other agent the same score.*

Proof. A state. Each good y_t moves to w_{t+1} only, and the goods h_{t+1} are distinct leftover goods; agents off the ring keep goods that are neither. The new holdings are a single good of the receiver or its pair. *Scores.* Along a want arrow, w_{t+1} receives a good it ranks above what it held. Along a pair arrow, w_{t+1} goes from its top to its pair. *Wanted goods.* Let g be wanted in Y, held alone by j. If j is off the ring, it keeps g. If $j = w_t$, then w_t is not free, so its arrow is a want arrow, and w_{t+1} now holds g alone. Lemma 4 gives validity. □

Lemma 6 (chain). *Let Y be valid, and x an agent with $Y_x = \{a_x\}$ and $b_x, c_x \in J$. Let $j_0 = x$, and as long as j_t holds a good that some agent wants, let j_{t+1} be the first such agent. If an agent repeats, the agents from*

its first occurrence on form a ring of want arrows. Otherwise the sequence stops at an agent j_k ($k \geq 0$) that holds nothing or a good nobody wants, and the following gives a valid state in which $x, j_1, \ldots, j_k$ have higher scores and every other agent the same score: x takes its pair; each j_t with $1 \leq t \leq k$ takes the good of j_{t-1}; the old good of j_k, if any, becomes leftover.

Proof. Agent x holds its top and wants nothing, so $j_t \neq x$ for $t \geq 1$. If $j_s = j_t$ with $1 \leq s < t$ is the first repeat, then $j_s, \ldots, j_{t-1}$ are distinct agents outside U, each holding a good that the next one wants: a ring of want arrows ($t - s \geq 2$, as no agent wants its own good). Otherwise: the goods b_x, c_x were leftover, and each moved good moves one step, so this is a state. Agent x goes from its top to its pair, and each j_t, $t \geq 1$, receives a good it wants. A wanted good is not b_x or c_x (they are leftover); if its holder is off the path it stays, and if its holder is j_t, then $t < k$ (the good of j_k is not wanted), and j_{t+1} now holds it alone. Lemma 4 gives validity. □

Applying a chain, or trading along a ring, is a *trade*. From now on, assume that no chain applies:

(NC) no agent x with $Y_x = \{a_x\}$ has both b_x and c_x in J.

Lemma 7 (the finishing test). *Let Y be a state with (NC), and o a free agent.*

(a) If o holds nothing, it has no blocker.
(b) If x is a blocker of o, then o holds a good y_o, and the pair of x is $\{y_o, h_x\}$ for a leftover good h_x.
(c) Let H_o be the set of the goods h_x of the blockers x of o. Then o can finish if and only if $|H_o| \leq |F| - 1$, and then it can finish with H_o. In particular a free agent holding nothing can finish.

Proof. Let x be a blocker of o: $b_x, c_x \in J \cup Y_o$, and not both are in J, by (NC). So one of them is in Y_o, which proves (a), and, as o holds at most one good, that good is y_o and the other one, h_x, is leftover: (b). A set $H \subseteq J$ contains b_x or c_x if and only if it contains h_x; so the sets H allowed by Definition 3 are those with $H_o \subseteq H \subseteq J$ and $|H| \leq |F| - 1$, which exist if and only if $|H_o| \leq |F| - 1$. If o holds nothing, $H_o = \emptyset$ and $|F| \geq 1$. □

Theorem 5 (Improvement Lemma). *Let Y be a valid state in which not every agent holds its pair and no free agent can finish. Then a chain*

or a ring gives a valid state in which every agent has at least its score and some agent a higher one.

Proof. If (NC) fails, Lemma 6 applies. Assume (NC). By Lemma 7, every free agent o holds a good and has $|H_o| \geq |F|$.

Distinct leftover goods. List the free agents in index order, $o_1, \ldots, o_f$. For $i = 1, \ldots, f$ in turn, let x_{o_i} be the first blocker of o_i whose leftover good is not among $q_{o_1}, \ldots, q_{o_{i-1}}$, and $q_{o_i} = h_{x_{o_i}}$. It exists because the blockers of o_i have $|H_{o_i}| \geq f > i - 1$ different leftover goods.

Pointers. For every agent $j \notin U$ let $\sigma(j) = x_j$ if j is free, and otherwise let $\sigma(j)$ be the first agent that wants the good of j (one exists, as $j \notin U \cup F$). In both cases $\sigma(j) \notin U$ and $\sigma(j) \neq j$.

The ring. Start at the first agent outside U and follow σ until an agent repeats. The agents from its first occurrence on form a cycle $w_0 \to \cdots \to w_{k-1} \to w_0$ with $k \geq 2$, each holding a single good (free agents by Lemma 7). The arrow from a free agent w_t is a pair arrow, with $h_{t+1} = q_{w_t}$, by Lemma 7(b); these goods are distinct for distinct free agents. Every other arrow is a want arrow. So the cycle is a ring, and Lemma 5 applies. □

Corollary 2 (Pareto-optimal states). *In the core case, in every valid state that no valid state Pareto-dominates in scores, either every agent holds its pair or some free agent can finish. In particular every core instance has a complete EFX$_0$ allocation in which all bundles but at most one have at most two goods.*

Proof. The first part is Theorem 5. Valid states exist (Lemma 3), there are finitely many, and one with the largest total score is Pareto-optimal; Theorem 4 finishes. □

6 The Algorithm Draft and Exchange

Algorithm 1 lists DE. It receives the values $v_i(g)$; ties are broken by index as in Section 2. Its loop runs on the core case left by peeling, with the notions of Sections 3–5.

Theorem 6 (DE is correct). *On every instance in which every agent values at most three goods,* DE *returns a complete EFX$_0$ allocation in which all bundles but at most one have at most two goods. It peels at most n agents and makes at most $4n$ trades.*

Algorithm 1 Draft and Exchange (DE)

Require: agents N, goods M, nonnegative values $v_i(g)$, every agent with $|R_i| \le 3$
$A \leftarrow N$, $G \leftarrow M$
while $|A| \ge 2$ and rule R1 applies to some agent of A with respect to G **do** ▷ peeling
    $i \leftarrow$ the first such agent; $X_i \leftarrow P_i$ (by rule R1); $A \leftarrow A \setminus \{i\}$, $G \leftarrow G \setminus P_i$
if $|A| = 1$ **then** give G to the agent of A and **return** X
for $i \in A$ in index order **do** ▷ the draft
    $Y_i \leftarrow$ the first of $\{a_i\}, \{b_i\}, \{c_i\}$ whose good is not taken, else nothing
loop
    if some x with $Y_x = \{a_x\}$ has $b_x, c_x \in J$ **then** ▷ chain
        apply the chain of the first such x, or the ring of want arrows it runs into (Lemma 6)
    else if every agent holds its pair **then** ▷ finish
        $o \leftarrow$ the first agent; $H \leftarrow \emptyset$; **break**
    else if some free agent o has $|H_o| \le |F| - 1$ **then** ▷ finish
        $o \leftarrow$ the first such agent; $H \leftarrow H_o$; **break**
    else ▷ ring
        trade along the ring of the proof of Theorem 5
give the k-th good of H (in the order of first blockers) to the k-th free agent other than o
$X_o \leftarrow Y_o \cup (J \setminus H)$; every other $i \in A$ keeps Y_i (plus its good of H, if any)
return X

Proof. Peeling. Each round removes one agent, so there are at most n rounds. Rule R1 for the agent i peeled with P_i is conditions (i) and (ii) of Lemma 1 on the instance left before it, so by induction from the last round down it suffices that DE returns an EFX$_0$ allocation of the agents A and goods G left at the end. If $|A| = 1$, the single agent faces no other bundle.

The core. Let $|A| \ge 2$. Rule R1 applies to no agent of A, so by Lemma 2 this is the core case. The draft gives a valid state (Lemma 3). A chain or a ring gives a valid state in which nobody's score drops and some score rises (Lemmas 6 and 5; the ring exists when it is used, by the proof of Theorem 5). The total score is an integer between 0 and $4|A|$, so there are at most $4|A| \le 4n$ trades, and the loop stops. When it stops, o can finish with H (Lemma 7(c); the chain test failed, so (NC) holds), or every agent holds its pair; Theorem 4 gives an EFX$_0$ allocation in which only X_o can have more than two goods. Peeled agents receive at most one good. □

A small example. Agents z, x, o, in index order, rank $g_4 \succ g_0 \succ g_1$, $g_0 \succ g_1 \succ g_2$ and $g_0 \succ g_4 \succ g_1$, with values 4, 3, 2; nobody values g_3. Nobody is peeled. In the draft z takes g_4, x takes g_0, and o takes g_1; the

leftover goods are g_2, g_3. Agent o wants g_0 and g_4, which are held alone: the state is valid, and o is the only free agent. No chain applies. Agent x is a blocker of o: if o took $\{g_1, g_2, g_3\}$, x would remove g_3 and see g_1 and g_2, worth $5 > 4$. So $H_o = \{g_2\}$, there is no other free agent to take g_2, and o cannot finish. The pointers are $\sigma(o) = x$, a pair arrow (x's pair is o's good g_1 and the leftover g_2), and $\sigma(x) = \sigma(z) = o$, want arrows ($o$ wants g_0 and g_4). Following them from z closes the ring $o \to x \to o$: x takes $\{g_1, g_2\}$ and o takes g_0. Now nobody wants anything, the free agents are z and o, and z has no blocker, so z finishes: $X_z = \{g_4, g_3\}$, $X_x = \{g_1, g_2\}$, $X_o = \{g_0\}$, an EFX_0 allocation.

Running time. An iteration of the loop needs the holder of each good, the goods each agent wants (at most three), the free agents, the blockers and their leftover goods, and a walk of at most n steps. Once the chain test has failed, (NC) holds, and an agent holding only its top is a blocker of at most one free agent (Lemma 7(b)), so all blockers and the sets H_o are found by one pass over the agents. With an array that records the holder of each good, an iteration takes $O(n + m)$ steps, and so does a round of peeling once the relevant goods of each agent are known. So DE runs in $O(n(n + m))$ steps, reading the input included. This count is machine-checked in Lean, with one unit per elementary operation, array read and array write (Section 8).

Real values. Every decision of DE compares one value of an agent with another, or with the sum of two of its values (rule R1, the rankings), and the proofs use nothing else about the values. So Theorems 1–3 hold for nonnegative real values.

Proof (of Theorems 1, 2 and 3). Theorem 6, the running time and the remark on real values. □

7 A Worked Example

On this instance every valid state that Pareto-dominates the draft state gives pairs to two agents at once, so a ring with a single pair arrow is not enough (Appendix A). Ten goods $g_0, \ldots, g_9$ and six agents, in index order x_1, x_1', x_2, x_2', o_1, o_2, with values 4, 3, 2 on their goods $a \succ b \succ c$ (any strictly balanced values with these rankings give the same run):

agent	$a \succ b \succ c$	after the draft	after the trade
x_1	$g_0 \succ g_4 \succ g_6$	$\{g_0\}$ (a)	$\{g_4, g_6\}$ (pair)
x'_1	$g_1 \succ g_4 \succ g_7$	$\{g_1\}$ (a)	$\{g_1\}$ (a)
x_2	$g_2 \succ g_5 \succ g_8$	$\{g_2\}$ (a)	$\{g_5, g_8\}$ (pair)
x'_2	$g_3 \succ g_5 \succ g_9$	$\{g_3\}$ (a)	$\{g_3\}$ (a)
o_1	$g_2 \succ g_3 \succ g_4$	$\{g_4\}$ (c)	$\{g_2\}$ (a)
o_2	$g_0 \succ g_1 \succ g_5$	$\{g_5\}$ (c)	$\{g_0\}$ (a)

The draft. Nobody is peeled. The four x's take their tops; o_1 finds g_2 and g_3 taken and takes g_4; o_2 takes g_5. Agent o_1 wants g_2 and g_3, o_2 wants g_0 and g_1, and the x's want nothing; the four wanted goods are held alone. The leftover goods are $g_6, \ldots, g_9$, and the free agents are o_1 and o_2. No chain applies: the b of each x is held by o_1 or o_2.

Nobody can finish. The blockers of o_1 are x_1 (pair $\{g_4, g_6\}$) and x'_1 (pair $\{g_4, g_7\}$), so $H_{o_1} = \{g_6, g_7\}$; symmetrically $H_{o_2} = \{g_8, g_9\}$. Both have $|H_o| = 2 > |F| - 1$. Concretely, if o_1 takes the leftovers, only o_2 can take one of g_6, g_7; with $H = \{g_6\}$, $X_{o_1} = \{g_4, g_7, g_8, g_9\}$, and x'_1, holding g_1 (worth 4), values X_{o_1} without g_8 at $3 + 2 = 5$. All ten candidate completions (finisher o_1 or o_2, with H empty or one leftover good) fail the EFX$_0$ test.

No small trade helps. Every want arrow ends at o_1 or o_2, so a ring with no pair arrow would have to leave one of them by a want arrow; but nobody wants their goods. A ring with exactly one pair arrow, from o_1 say, would return to o_1 by want arrows; but x_1 and x'_1 have want arrows only to o_2, and the arrows from o_2 are pair arrows. In fact exactly four valid states Pareto-dominate the draft state, and each gives pairs to two agents.

The trade. DE takes the first blocker of each free agent: $x_{o_1} = x_1$, with $q_{o_1} = g_6$, and $x_{o_2} = x_2$, with $q_{o_2} = g_8$. So $\sigma(o_1) = x_1$, $\sigma(x_1) = o_2$ (o_2 wants g_0), $\sigma(o_2) = x_2$ and $\sigma(x_2) = o_1$ (o_1 wants g_2): the ring of Figure 1. Along it, x_1 takes $\{g_4, g_6\}$, o_2 takes g_0, x_2 takes $\{g_5, g_8\}$ and o_1 takes g_2. The total score rises from 14 to 20.

Finishing. Now nobody wants anything, the leftover goods are g_7 and g_9, and the free agents are x'_1, x'_2, o_1, o_2. No chain applies (b of x'_1 is g_4, now held by x_1; similarly for the others). The first free agent, x'_1, has no blocker: no other agent holding its top has both b and c in $\{g_1, g_7, g_9\}$. So

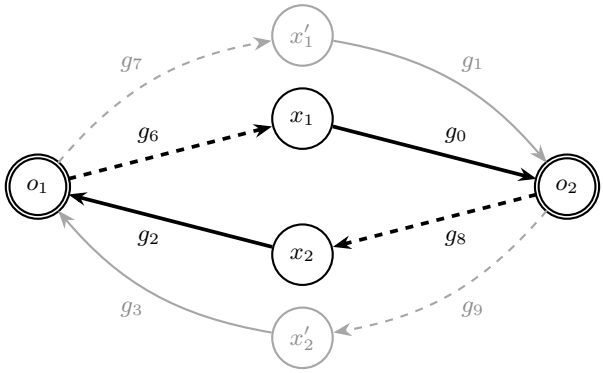


Fig. 1. The arrows of the draft state of Section 7. Double circles are the free agents. Dashed arrows are pair arrows, labelled by the blocker's leftover good; solid arrows are want arrows, labelled by the good that moves. The ring chosen by DE is drawn in black. Every cycle of these arrows has four agents and two pair arrows.

x'_1 finishes with $H = \emptyset$, and DE returns

$$X_{x_1} = \{g_4, g_6\}, \quad X_{x'_1} = \{g_1, g_7, g_9\}, \quad X_{x_2} = \{g_5, g_8\},$$
$$X_{x'_2} = \{g_3\}, \quad X_{o_1} = \{g_2\}, \quad X_{o_2} = \{g_0\}.$$

Directly: the only bundle of three goods, $X_{x'_1}$, is worth 3 to o_2 (only g_1), which holds g_0, worth 4, and 2 to x'_2 (only g_9), which holds g_3, worth 4; after removing one good, a bundle of two goods is worth at most 3 to any other agent, and every agent holds at least 4.

8 Verification

Lean. The proofs of Sections 2–6 are machine-checked in core Lean 4 (`lean/EFX/K3DE*.lean` in [11], and `K3Extras.lean` for serial dictatorship): no library beyond Lean's own, no unproved step, every declaration on Lean's three standard axioms, checked by the repository's `lean/check.sh`, which also replays the library in a fresh kernel. DE is a Lean function, and the main theorem states that on every instance with natural-number values in which every agent positively values at most three goods it returns an EFX_0 allocation in which all bundles but at most one have at most two goods, after at most n peeling rounds and $4n$ trades. The Lean files use other names for some notions: *needs* for wants, *junk* for leftover goods, *utility* for score, *exposed* for blocker, *protecting good* for the leftover good h_x, *need arcs* and *exposure arcs* for want and pair arrows, *need cycle* and *exchange cycle* for rings, *pair chain* for a

chain, (P) for (NC), and *valid absorber* for an agent that can finish (a notion that also allows a pair holder to finish, as when every agent holds its pair). One Lean lemma covers chains and rings (a chain is closed by an arrow from its last agent back to x).

Further files check the rest: Corollary 1; DE on ordered values such as the nonnegative reals (it runs on natural-number values computed from the given ones with at most $n(m + 12)$ comparisons, which answer every comparison of two subset sums of an agent as the given values do, so its output is EFX_0 for the given values); the facts of Section 7 and Figure 1 (by evaluation); Example 1; and the results of the appendices. The running time is checked on a counted program, proved to return exactly DE's output: at most $750(n + 1)(n + m + 1)$ units, reading the input included, with one unit per elementary operation, comparison, array read and array write, as in a random-access machine, arrays being used single-threaded. That the units match a machine is a property of the counted program, checked by reading it. The formal statements were written by AI agent sessions of the coding assistant that helped with this work (Claude Code) and compared with the text by independent AI sessions (`k3/simplify/po/referee/lean\_audit*.md` in [11]).

Tests. The tests are evidence, not proof. Three implementations, each asserting every claim of a proof and checking outputs against the raw definition of EFX_0 with exact arithmetic, ran without failure on every profile with $n = 2$ and $m \leq 6$, with $n = 3$ and $m \leq 7$, and with $n = 4$ and $m \leq 6$ (in part), on cores with $n = 5, 6$ and on random profiles with up to 9 agents: about 1.7 million valid states for the proof of Section 5, about 15.5 million for an independently found second proof (a walk instead of a choice of distinct leftover goods), and about 2.9 million runs of a referee's implementation written from the text alone (`k3/simplify/po/` in [11]). The first and the referee's run DE's loop with the same three steps (up to the choice among leftover goods); the second tests the lemmas of Sections 4 and 5 on its own loop, which lets a free agent holding nothing finish first. The script `paper/k3-simple/examples/check\_examples.py`, written from this paper with the loop as stated here, recomputes every example and number of the paper and runs DE, peeling included, on random instances with ties, unbalanced agents and goods nobody values.

9 Conclusion and Open Problems

Every instance in which each agent positively values at most three goods has a complete EFX_0 allocation with at most one bundle of more than

two goods, and DE computes one with at most n peeling steps and $4n$ trades. The proof rests on one fact: if no free agent can take the leftover goods, then every agent can point at an agent that would gladly take its good, the free agents at blockers with distinct leftover goods, and the pointers close a ring along which everyone gains.

Open problems. (1) EFX_0 with four relevant goods per agent (for ordinary EFX, see [16]). With four goods an agent can hold one, two or three of them, and the order of holdings is not determined by the ranking alone, so the scores of Section 3 have no direct analogue. (2) A better bound than $4n$ trades (at most 8 were needed on random instances; from the draft, the rings of Appendix A of depth 4 need up to 12), or a draft that needs none. (3) The least size of the large bundle; Proposition 2(b) shows that three goods do not always suffice.

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A How Large Must a Ring Be?

Call a *short move* a chain or a ring with at most one pair arrow. A natural simpler algorithm would use only short moves. The next proposition shows that the instance of Section 7 is as small as possible in its number of agents among states where short moves are not enough.

Proposition 1 (when short moves are not enough). *In the core case, let Y be a valid state in which not every agent holds its pair, no free agent can finish, and no short move applies. Then $|F| \geq 2$, $n \geq 6$ and $m \geq 8$; if $|F| \geq 3$, then $n \geq 9$ and $m \geq 12$. The bounds $n = 6$ and $m = 8$ are attained.*

Proof. No chain applies, so (NC) holds, and by Lemma 7 every free agent o holds a good and has $|H_o| \geq |F|$, hence at least $|F|$ blockers. The blockers of two different free agents o, o' are different: a common blocker would have the pair $\{y_o, y_{o'}\}$, with no leftover good, against Lemma 7(b). There is no ring of want arrows, so a walk that repeatedly moves to an agent that wants the current agent's good never repeats an agent; it ends at an agent whose good nobody wants, or that holds nothing, which is free. Every agent holds a good: pair holders two, free agents one by Lemma 7, others a wanted good. Let $k = |F|$; $k \geq 1$, because the walk from an agent outside U ends at a free agent.

If $k = 1$, let $F = \{o\}$ and x a blocker of o. Then x is not free, and its walk ends at o; with the pair arrow $o \to x$ this is a ring with one pair arrow, a short move. So $k \geq 2$.

If $k = 2$, let $F = \{o_1, o_2\}$. Agent o_1 has at least two blockers and only o_2 is another free agent, so some blocker x of o_1 is not free. Its walk cannot end at o_1, as above, so it ends at o_2, entering it by a want arrow ($x \neq o_2$). So o_2 wants a good and does not hold its top: o_2 is not a blocker of o_1, and no blocker of o_1 is free. The same holds for o_2. So $n \geq 2 + 2 + 2 = 6$, and $m \geq n + |J| \geq 6 + 2 = 8$.

If $k \geq 3$, the blockers of the k free agents are at least k^2 different agents, at most k of them free, so $n \geq |F| + (k^2 - k) = k^2 \geq 9$, and $m \geq n + |J| \geq k^2 + k \geq 12$.

Attained. The instance of Section 7 has $n = 6$ ($m = 10$). For $m = 8$: o_1 ranks $g_4 \succ g_5 \succ g_0$, o_2 ranks $g_6 \succ g_7 \succ g_1$, x_1 ranks $g_4 \succ g_1 \succ g_2$, x_1' ranks $g_5 \succ g_3 \succ g_1$, x_2 ranks $g_6 \succ g_0 \succ g_2$, x_2' ranks $g_7 \succ g_3 \succ g_0$, with o_1 holding g_0, o_2 holding g_1 and every x its top. Then $J = \{g_2, g_3\}$, $F = \{o_1, o_2\}$, the blockers of o_2 are x_1, x_1' (leftover goods g_2, g_3) and those of o_1 are x_2, x_2' (leftover goods g_2, g_3); nobody can finish, and no short move applies. □

In the second instance the distinct leftover goods matter: of the four cycles $o_2 \to x \to o_1 \to x' \to o_2$ with two pair arrows, two use the same leftover good twice, and trading along them would give it to two agents; the other two are rings.

No bounded number of pair arrows suffices. For every $k \geq 2$ and $d \geq 1$ there are valid states with k free agents $o_0, \ldots, o_{k-1}$ in which each o_i is the root of a binary tree of depth d whose agents want the goods of their two children, and the leaves of the tree of o_{i+1} hold their tops and are blockers of o_i (indices modulo k), with leftover goods that are distinct within each tree: one of its own per leaf, or the goods of a common pool

of k, the leaves of each tree taking them in turn. Every cycle of want and pair arrows there has exactly k pair arrows, and no free agent can finish if and only if $k \leq 2^d$, the number of leaves of a tree. For $k = 2$ and $d = 1$ this is the instance of Section 7. We omit the proofs of these two facts; both are machine-checked in Lean for every k and d.

B Limits of the Shape

Could every instance have an EFX$_0$ allocation with all bundles of at most two goods, or with one bundle of exactly three? No. Write $\theta_i(B)$, the *threat* of B to i, for the largest value of $v_i(B \setminus \{h\})$ over $h \in B$ (0 if $B = \emptyset$): agent i does not envy X_j in the sense of EFX$_0$ if and only if $v_i(X_i) \geq \theta_i(X_j)$, and i is *safe* if this holds for every $j \neq i$. By additivity, $\theta_i(B) = 0$ if $|B| \leq 1$; $\theta_i(B) \leq v_i(B \cap R_i)$, with equality if B has a good outside R_i; and $\theta_i(\{g, h\}) = \max(v_i(g), v_i(h))$.

Lemma 8 (cases of safety). *Let agent i have $R_i = \{a, b, c\}$ with $v_i(a) > v_i(b) > v_i(c) > 0$ and $v_i(a) < v_i(b) + v_i(c)$, and let X be an allocation. Then i is safe in X if and only if one of the following holds:*

(T) *i holds a, and either i also holds b or c, or b and c are not together in a bundle of at least three goods;*
(BC) *i holds b and c;*
(B) *i holds b, and a is alone;*
(C) *i holds c, and a and b are alone;*
(E) *a, b and c are all alone.*

Proof. Write a, b, c also for the values. The sums of subsets of $\{a, b, c\}$ are ordered as $0 < c < b < a < b + c < a + c < a + b < a + b + c$. Let $S = X_i \cap R_i$.

- $S \supseteq \{a, b\}$ or $S \supseteq \{a, c\}$: every other bundle meets R_i in at most one good, so its threat is at most $b < a + c$. Safe: (T).
- $S = \{b, c\}$: every threat is at most $a < b + c$. Safe: (BC).
- $S = \{a\}$: if b, c are in different bundles, each threat is at most $b < a$; if $\{b, c\}$ is a bundle, its threat is $b < a$; if a bundle strictly contains $\{b, c\}$, it has a third good, outside R_i, and its threat is $b + c > a$. So: (T).
- $S = \{b\}$: if a is not alone, its bundle has threat at least $a > b$; if a is alone, every threat is at most $c < b$. So: (B).
- $S = \{c\}$: as before a must be alone, and if b is not alone its bundle has threat at least $b > c$. So: (C).

– $S = \emptyset$: a bundle with a good of R_i and at least two goods has a positive threat. So: (E).

Conversely, each of (T), (BC), (B), (C), (E) describes safe situations above: (T) is the first and third items; (BC) is $S = \{b, c\}$ or $S = \{a, b, c\}$; (B) forces $a \notin S$, so $S = \{b\}$ with a alone, or $S = \{b, c\}$; (C) forces $S = \{c\}$ with a and b alone; and (E) means that S is empty or a single alone good, which is (T), (B) or (C). □

Proposition 2 (limits of the shape). *Take values with* $v_i(a_i) > v_i(b_i) > v_i(c_i) > 0$ *and* $v_i(a_i) < v_i(b_i) + v_i(c_i)$ *for every agent.*

(a) Let agents 0, 1, 2 *rank* $g_0 \succ g_1 \succ p_i$*, where* p_i *is valued only by agent* i*. No EFX*$_0$ *allocation has all bundles of at most two goods, and* $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$*, to the three agents in any order, is EFX*$_0$*.*

(b) Let agents 0, 1, 2, 3 *rank* $g_0 \succ g_2 \succ p_0$*,* $g_0 \succ g_2 \succ p_1$*,* $g_1 \succ g_2 \succ p_2$ *and* $g_1 \succ g_2 \succ p_3$*, with* p_i *valued only by agent* i*. Every EFX*$_0$ *allocation has bundle sizes* $4, 1, 1, 1$*, and one exists.*

Proof. (a) Five goods in three bundles of at most two goods have sizes $2, 2, 1$, so exactly one good is alone. Two agents do not hold their top g_0: case (T) fails for them, and (C) and (E) need two or three alone goods. So both are in case (BC) or (B), and both hold g_1, which is impossible. In $\{g_0\}, \{g_1\}, \{p_0, p_1, p_2\}$ the three agents are in cases (T), (B) and (C).

(b) Each of the tops g_0, g_1 has an agent ranking it first that does not hold it, a *loser*; the two losers differ. A loser is safe only by (BC) or (B), which need it to hold g_2, or by (C) or (E), which need g_2 alone. The two losers cannot both hold g_2, so g_2 is alone, and nobody is in case (BC). A loser holding $\{g_2\}$ needs its top alone; a loser not holding g_2 needs its top and g_2 alone. So g_0, g_1, g_2 are alone, held by three agents, and the fourth bundle is $\{p_0, p_1, p_2, p_3\}$. Such an allocation exists: g_0 to agent 0, g_1 to agent 2, g_2 to agent 1, the four goods p_i to agent 3 (cases T, B, T, C). □

In (a) every agent values exactly three goods and is strictly balanced, so the one large bundle of Theorem 2 cannot be avoided; by (b), its size cannot be bounded by three.

C Related Work in a Table

Table 1. Existence results for complete allocations under restrictions. *Version cited*: the published version where one exists, else arXiv. [a]Goods nobody values go to an arbitrary agent. [b]Includes additive. [c]Extended abstract with a proof sketch. [d]An AAMAS 2025 extended abstract has the degree and girth conditions. [e]Accepted at EC 2026.

Result	Restriction (agents, goods, who values what)	Notion	Complete	Valuations	Algorithm	Version cited
[6]	$n = 3$	EFX_0	yes	additive	pseudo-polynomial	JACM 2024
[8]	at most three distinct valuations	EFX_0	yes	additive	no bound stated	EC 2025
[14]	$m \leq n + 3$	EFX_0	yes	monotone	none stated	MOR 2024
[3]	$n = 4$, $m \leq 9$	EFX_0	yes	additive	none stated	arXiv
[5]	$m = n + 4$	EFX_0	yes, claimed	additive	none stated	Zenodo, not peer-reviewed
[7]	graphs: at most two valuers per good, two agents share at most one good	EFX_0	yes	monotone	polynomial time	MOR 2026 (EC 2023)
[1]	multigraphs: at most two valuers per good	EFX_0	yes	cancelable[b]	polynomial time	arXiv[e]
[15]	multigraphs: bipartite, or at most $\lceil n/4 \rceil - 1$ neighbours per agent, or girth at least six (non-parallel edges)	EFX_0	yes	monotone	not efficient (stated)	arXiv[d]
[9]	at most two valuers per good, $v_i(g) \in \{0, v_g\}$	EFX_0	yes	restricted additive	no time bound stated	WINE 2024
[10]	hypergraphs of girth at least four (a good joins its valuers)	EFX_0	yes	monotone	polynomial time	AAAI 2026
[16]	at most four relevant goods per agent	EFX	yes[a]	additive	polynomial time, announced	AAMAS 2024[c]
[12]	at most two relevant goods per agent	EFX_0	yes	additive; monotone	serial dictatorship; Lean-checked	GitHub, v1.2.0
this paper	at most three relevant goods per agent; any number of valuers per good	EFX_0	yes	additive	polynomial time, at most $4n$ trades	this paper